

Digitisation

This is the process of converting analogue signals to digital signals.

base among niche group of hobbyists, notably the Boy Scouts of America. Radio technology continued to improve through the 1920s

into the 1930s with the improvement of vacuum tube, the invention of the early ancestors of diodes, etc. Some of the major contributors to these achievements was Westinghouse laboratories, based in the USA and, as always, Marconi.

The next great leap in the field of radio came in 1933 with the development of FM. This revolutionary technology insulated the signal from external electronic interference and allowed the transmission of crystal clear audio and other signals across radio waves. However its technological features limited its usage to short range (a city wide, for example) applications.

With the end of World War 2, radio stations and devices spread across Europe and the rest of the world. Soon, radio became a commonplace device. The 1950s witnessed the rapid miniaturisation of radio receivers thanks to the discovery of transistors and diodes. Over the next 20 years, transistors replaced vacuum tubes in all applications except the most specialised.

The 1960s witnessed a new revolution in wireless communications with the advent of communication satellites. With the launch of Telstar – the first communications only satellite, it became possible to transmit across the world, beyond the line of sight. Communication satellites stay in geostationary orbit.

The late 1960s also witnessed the digitisation of radios, mainly in long distance telephone networks. The 1970s saw the advent of radio and satellite navigation systems, originating from attempts by the US navy to precisely navigate their ships. In 1987, the GPS system of satellites was launched.

The 1990s witnessed the birth of various technologies merging computers and other devices such as mobile phones, PDAs and wireless technologies. Wireless LAN, Bluetooth, etc., are the offspring of this revolution.

Geostationary orbit

An orbit where one revolution by the satellite around the Earth is 24 hours, so that it remains directly above one point on the Earth at all times.